13.8 OBLIQUE ROTATION

Orthogonal rotations are appropriate for a factor model in which the common factors are assumed to be independent. Many investigators in social sciences consider oblique (non-orthogonal) rotations, as well as orthogonal rotations. The former rotations are often suggested after one view of the estimated factor loadings that do not follow the assumption made in the factor model. Nevertheless, an oblique rotation is frequently a useful aid in factor analysis.

In other words, oblique rotation is suitable when the factors are considered to be correlated. Oblique rotation is more complex than orthogonal rotation, since it can involve one of two coordinate systems, a system of primary axes or a system of reference axes. Additionally, oblique rotation produces a pattern matrix that contains the factor or item loadings and factor correlation matrix that includes the correlations between the factors. The commonly used oblique rotation techniques are Direct Oblimin and Promax. Direct Oblimin attempts to simplify the structure and the mathematics of the output of the problem under consideration, while Promax is used because of its speed in larger datasets. Promax involves raising the loadings to a power of four which ultimately results in greater correlations among the factors and achieves a simple structure.

13.9 FACTOR SCORES

A factor score can be considered to be a variable describing how much an individual would score on a factor. One of the methods to produce factor score is called Bartlett method (or regression approach) which produces unbiased scores that are correlated only with their own factor. Another method is called the Anderson-Rubin method which produces scores that are uncorrelated and standardized. In factor analysis, interest is usually centered on the parameters in the factor model. However the estimated value of the common factors, called factor scores, may also be required. These quantities are often used for diagnostic purposes, as well as inputs to a subsequent analysis.

Factor scores are not estimates of unknown parameters in the usual sense. Rather, they are estimates of values for the unobserved random factors F_j, j = 1, 2,……,m. We try to find out the vectors of m unobserved common factors that underlie our model to estimate those factors. Therefore, for the factor model

$$X - \mu = L\,F + \varepsilon$$

We may wish to estimate the vectors of the factor scores F_1, F_2,…….,F_m.

13.10 METHODS FOR ESTIMATING FACTOR SCORE

A number of different methods are used for estimating factor scores from the data. These include

1) Ordinary Least Square Method
2) Weighted Least Square Method
3) Regression Method

Now, the above methods are discussed below in brief.

13.10.1 Ordinary Least Squares Method

If the factor loadings are estimated by the principal component method, it is customary to generate factor scores using an unweighted (ordinary) least squares procedure. The L's are factor loading and the f are unobserved common factors. The vector of common factors for subject i, is found by minimizing the sum of the squared residuals:

$$\sum_{j=1}^{p} \varepsilon_{ij}^2 = \sum_{j=1}^{p} \left(X_{ij} - \mu_j - l_{j1}F_1 - l_{j2}F_2 - \cdots - l_{jm}F_m\right)^2 = (X - \mu - L\,F)'(X - \mu - L\,F)$$

This is like a least squares regression, except in this case we already have estimates of the parameters (the factor loadings), but we wish to estimate the explanatory common factors. In matrix notation the solution is expressed as:

$$F_j = (L'L)^{-1}\, L'\left(X_j - \mu\right) \qquad (13.16)$$

In practice, the parameters are unknown in nature so we take the estimates of the factor loadings as follows:

$$\hat{F}_j = \left(\hat{L}'\,\hat{L}\right)^{-1}\hat{L}'\left(X_j - \overline{X}\right)$$

or

$$\hat{F}_j = \left(\hat{L}_z'\,\hat{L}_z\right)^{-1}\hat{L}_z'\, z_j \qquad (13.17)$$

For the standardized data, since L= $\left[\sqrt{\hat{\lambda}_1}e_1 \vdots \sqrt{\hat{\lambda}_2}e_2 \vdots \cdots \vdots \sqrt{\hat{\lambda}_m}e_m\right]$, we have

$$\hat{F}_j = \begin{bmatrix} \dfrac{1}{\sqrt{\hat{\lambda}_1}}\,\hat{e}_1'\left(X_j - \overline{X}\right) \\ \dfrac{1}{\sqrt{\hat{\lambda}_2}}\,\hat{e}_2'\left(X_j - \overline{X}\right) \\ \vdots \\ \dfrac{1}{\sqrt{\hat{\lambda}_m}}\,\hat{e}_m'\left(X_j - \overline{X}\right) \end{bmatrix}$$

We see that the $\hat{F}_j$ are approximately same as that of obtained in the principal component method with the unrotated factor loadings. Here, we have for the factor scores

$$\frac{1}{n}\sum_{j=1}^{n} \hat{F}_j = 0 \qquad \text{(sample mean)}$$

$$\frac{1}{n-1}\sum_{j=1}^{n} \hat{F}_j\hat{F}_j' = I \qquad \text{(sample variance)}$$

13.10.2 Weighted Least Squares Method

The difference between weighted least square and the ordinary least square is that the squared residuals are being divided by the specific variances as shown below. Let us suppose that the mean vector μ, the factor loading L, and the specific variance Ψ are known for the factor model

$$X - \mu = L\,F + \varepsilon$$

Further, regard the specific factors $\varepsilon' = [\varepsilon_1, \varepsilon_2, \dots\dots, \varepsilon_p]$ as errors. Since, $\mathrm{Var}(\varepsilon_i) = \Psi_i$; i= 1, 2,……,p, need not be equal. Bartlett [2] has suggested that weighted least squares be used to estimate the common factor values.

The sum of squares of the errors, weighted by the reciprocal of their variance is

$$\sum_{i=1}^{p} \frac{\varepsilon_i^2}{\Psi_i} = \sum_{j=1}^{p} \varepsilon' \Psi^{-1} \varepsilon = (X - \mu - LF)' \Psi^{-1} (X - \mu - LF) \tag{13.18}$$

The solution is given by the expression where Ψ is the diagonal matrix whose diagonal elements are equal to the specific variances.

$$\hat{F} = \left(L' \Psi^{-1} L\right)^{-1} L' \Psi^{-1} (X - \mu)$$

We take the estimates $\hat{L}, \hat{\Psi}$ and $\hat{\mu} = \overline{X}$ as the true values and obtain the factor scores for the j^{th} case as

$$\hat{F}_j = \left(\hat{L}' \hat{\Psi}^{-1} \hat{L}\right)^{-1} \hat{L}' \hat{\Psi}^{-1} \left(X_j - \overline{X}\right) \tag{13.19}$$

When $\hat{L}$ and $\hat{\Psi}$ are determined by the maximum likelihood method, these estimates must satisfy the uniqueness condition, $\hat{L}' \hat{\Psi}^{-1} \hat{L} = \hat{\Delta}$, a diagonal matrix.

13.10.3 Regression Method

This method is used when you are calculating maximum likelihood estimates of factor loadings. A vector of the observed data, supplemented by the vector of factor loadings for the ith subject, is considered.

In factor model $X - \mu = L\,F + \varepsilon$, we initially treat the loadings matrix L and specific variance matrix Ψ is known. When the common factors F and specific factors ε are jointly distributed with means and covariances, the linear combination $X - \mu = L\,F + \varepsilon$ has an $N_p(0, LL' + \Psi)$ distribution. Moreover, the joint distribution of $(X - \mu)$ and F is $N_{m+p}(0, \Sigma^*)$, where

$$\Sigma^* = \begin{bmatrix} \Sigma = LL' + \Psi & \vdots & L \\ \dots\dots\dots\dots\dots\dots & \vdots & \dots\dots \\ L' & \vdots & I \end{bmatrix} \tag{13.20}$$

and 0 is an (m×p)×1 vector of zeros. We find that the conditional distribution of $(F|x)$ is multivariate normal with

$$\text{mean} = E(F|x) = L' \Sigma^{-1} (x - \mu) = L'(LL' + \Psi)^{-1} (x - \mu) \tag{13.21}$$

and

$$\text{Covariance} = E(F|x) = I - L^{\Sigma^{-1}} L = I - L'(LL' + \Psi)^{-1} L \tag{13.22}$$

The quantities $L'(LL' + \Psi)^{-1}$ are the coefficients in a multivariate regression of the factors on the variables. Estimates of these coefficients produce factor scores that are analogous to the estimates of the conditional mean values in multivariate regression analysis. Consequently, given any vector of observations x_j, and taking the maximum likelihood estimates $\hat{L}$ and $\hat{\Psi}$ as the true values, we see that the j^{th} factor score vector is given by

$$\hat{f}_j = \hat{L}'\hat{\Sigma}^{-1}(x_j - \bar{x}) = \hat{L}'(\hat{L}\hat{L}' + \hat{\Psi})^{-1}(x_j - \bar{x}); \qquad j = 1,2,\ldots\ldots n \qquad (13.23)$$

The calculation of $\hat{f}_j$ in equation (13.23) can be simplified by using the matrix identity

$$\hat{L}'(\hat{L}\hat{L}' + \hat{\Psi})^{-1} = \left(I + \hat{L}'\hat{\Psi}^{-1}\hat{L}\right)^{-1}\hat{L}'\hat{\Psi}^{-1} \qquad (13.24)$$

This identity allows us to compare the factor scores in equation (13.23), generated by the regression argument, with those generated by the weighted least squares method in equation (13.19). Temporarily, If we denote the former by $\hat{F}_j^R$ and the latter by $\hat{F}_j^{LS}$. Then by using equation ( ), we have

$$\hat{F}_j^{LS} = \left(\hat{L}'\hat{\Psi}^{-1}\hat{L}\right)^{-1}\left(I + \hat{L}'\hat{\Psi}^{-1}\hat{L}\right)\hat{F}_j^R = \left(I + \left(\hat{L}'\hat{\Psi}^{-1}\hat{L}\right)^{-1}\right)\hat{F}_j^R \qquad (13.25)$$

For maximum likelihood estimates $\left(\hat{L}'\hat{\Psi}^{-1}\hat{L}\right)^{-1} = \hat{\Delta}^{-1}$ and if the elements of this diagonal matrix are close to zero, the regression and generalized least squares methods will give nearly the same factor scores.

There may be made an attempt to reduce the effect of a incorrect determination of the number of factors, to calculate the factor scores in equation (13.23) by using S (the sample covariance matrix) instead of $\hat{\Sigma} = \hat{L}\hat{L}' + \hat{\Psi}$. Then we have the following results:

$$\hat{f}_j = \hat{L}'S^{-1}(x_j - \bar{x}); \qquad j = 1, 2, \ldots\ldots n \qquad (13.26)$$

or, if a correlation matrix is factored

$$\hat{f}_j = \hat{L}_z' R^{-1} z_j; \qquad j = 1, 2, \ldots\ldots n \qquad (13.27)$$

where

$$z_j = D^{-1/2}(x_j - \bar{x}) \quad \text{and} \quad \hat{\rho} = \hat{L}_z\hat{L}_z' + \hat{\Psi}_z$$

Again, if rotated loadings $\hat{L}^* = \hat{L}\,T$ are used in place of the original loadings in eq (13.26), the subsequent factor scores $\hat{f}_j^*$ are related to $\hat{f}_j$ by

$$\hat{f}_j^* = T'\hat{f}_j; \qquad j = 1, 2, \ldots\ldots, n$$

A numerical measure between the factor scores generated from two different calculation methods is provided by the sample correlation coefficient between scores on the same factor.

Source: Adapted from Applied multivariate statistical analysis by Johnson R.A. and Wichern D.W. (2002)

Check Your Progress 2

1) The correlation matrix for chicken- bone measurements is

$$\begin{bmatrix} 1.000 & & & & & \\ .505 & 1.000 & & & & \\ .569 & .422 & 1.000 & & & \\ .602 & .467 & .926 & 1.000 & & \\ .621 & .482 & .877 & .874 & 1.000 & \\ .603 & .450 & .878 & .894 & .937 & 1.000 \end{bmatrix}$$

The following estimated factor loadings were extracted by the maximum likelihood procedure:

Variable	Estimated factor loadings		Varimax rotated estimated factor loadings	
	F_1	F_2	F_1^*	F_1^*
Skull length	.602	.200	.484	.411
Skull breadth	.467	.154	.375	.319
Femur length	.926	.143	.603	.717
Tibia length	1.000	.000	.519	.855
Humerus length	.874	.476	.861	.499
Ulna length	.894	.327	.744	.594

Using the unrotated estimated factor loadings, obtain the maximum likelihood estimates of the following:

a) The specific variances

b) The communalities

c) The proportion of variance explained by each factor

d) The residual matrix $R - \hat{L}_z \hat{L}_z' - \hat{\Psi}_z$

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1) From the above exercise 1, Compute the value of the varimax criterion using both unrotated and rotated estimated factor loadings Also interpret the result.

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2) The following table provides data of the Students entering in a certain MBA program must take three required courses in Finance, Marketing and Business policy. Let X_1, X_2, and X_3, respectively, represent a student's grades in these courses. The available data consist of the grades of five students (in a 10-point numerical scale above the passing mark), as shown in Table 13.1. Using an appropriate statistical package (SPSS, Minitab, SAS etc) run a factor analysis and interpret the results.

Table: 13.1: Students Grade

Grade in (→) Student No.(↓)	Finance (X_1)	Marketing (X_2)	Business Policy (X_3)
1	3	6	5
2	7	3	3
3	10	9	8
4	3	9	7
5	10	6	5

(Example adapted from Peter Tryfos, 1997, version printed: 14-3-2001).

13.11 LET US SUM UP

Factor analysis is something like an art. Factor analysis is used to study the patterns of relationship among many dependent variables to have an idea about the nature of the independent variables not measured directly.

Factor analysis is used to identify latent constructs or factors. It is commonly used to reduce variables into a smaller set factors to save time and facilitate easier interpretations. There are many extraction techniques such as Principal component method and Maximum Likelihood. The factor analysis and the principal component analysis are among the oldest of the multivariate statistical methods of data reduction. Mathematically, factor analysis is complex and the criteria used to determine the number and significance of factors are vast. There are two types of rotation techniques – orthogonal rotation and oblique rotation. Orthogonal rotation (e.g., Varimax and Quartimax) involves uncorrelated factors whereas oblique rotation (e.g., Direct Oblimin and Promax) involves correlated factors. The interpretation of factor analysis is based on rotated factor loadings, rotated eigenvalues, and scree test. In reality, investigators often use more than one extraction and rotation technique based on pragmatic reasoning rather than theoretical reasoning.

13.12 KEY WORDS

Communality : It shows the variance of an observed variable accounted for by the common factors; in an orthogonal factor model. It is equivalent to the sum of the squared factor loadings.

Common Factor : It implies the unmeasured (or hypothetical) underlying variable which is the source of variation in at least two observed variables under consideration.

Confirmatory Factor Analysis (CFA) : This technique of the factor analysis attempts to confirm hypotheses and uses path analysis diagrams to represent variables and factors.

Exploratory Factor Analysis (EFA) : The Exploratory Factor Analysis technique tries to uncover complex patterns by exploring the dataset and testing predictions.

Diagonal Matrix	:	It is a square matrix in which the entries outside the main diagonal are all zero.
Eigenvalue (characteristic root)	:	It is a mathematical property of a matrix; used in relation to the decomposition of a covariance matrix, both as a criterion of determining the number of factors to extract and a measure of variance accounted for by a given dimension.
Eigenvector	:	It is a vector associated with its respective eigenvalue; obtained in the process of initial factoring; when these vectors are appropriately standardized, they become factor loadings.
Factor Extraction	:	It is the initial stage of factor analysis in which the covariance matrix is resolved into a smaller number of underlying factors or components.
Factors	:	It implies hypothesized, unmeasured, and underlying variables which are presumed to be the sources of the observed variables; often divided into unique and common factors.
Factor Loading	:	It is a general term referring to a coefficient in a factor pattern or structure matrix.
Factor Pattern Matrix	:	It refers to a matrix of coefficients where the columns usually refer to common factors and the rows to the observed variables; elements of the matrix represent regression weights for the common factors where an observed variable is assumed to be a linear combination of the factors; for an orthogonal solution, the pattern matrix is equivalent to correlations between factors and variables.
Kaiser criterion	:	In this criterion, first we can retain only factors with eigenvalues greater than 1 when factors are calculated from correlation matrix. In essence this is like saying that, unless a factor extracts at least as much as the equivalent of one original variable, we drop it. This criterion was proposed by Kaiser (1960), and is probably the one most widely used.
Orthogonal Factors	:	It indicates the factors that are not correlated with each other; factors obtained through orthogonal rotation.
Orthogonal Rotation	:	It refers to the operation through which a simple structure is sought under the restriction that factors be orthogonal (or uncorrelated); factors obtained through this rotation are by definition uncorrelated.
Principal Axis	:	It is a method of initial factoring in which the

Factoring adjusted correlation matrix is decomposed hierarchically; a principal axis factor analysis with iterated commonalities leads to a least squares solutions of initial factoring.

Principal Components : It reflects linear combinations of observed variables, possessing properties such as being orthogonal to each other, and the first principal component representing the largest amount of variance in the data, the second representing the second largest and so on; often considered variants of common factors, but more accurately they are contrasted with common factors which are hypothetical.

Scree Test : It is a rule of thumb criterion for determining the number of significant factors to retain; it is based on the graph of roots (eigenvalues); claimed to be appropriate in handling disturbances due to minor (unarticulated) factors.

Varimax : It is a method of orthogonal rotation which simplifies the factor structure by maximizing the variance of a column of the pattern matrix.

13.13 SOME USEFUL BOOKS

1) Anderson, T. W. (1984); *An Introduction to Multivariate Statistical Analysis*, John Wiley & Sons, Second Edition.

2) Bartlett, M.S. (1954); A note on Multiplying factors for various Chi-squared Approximations, Journal of the Royal Statistical Society, 16, 296-298.

3) Johnson R.A., & Wichern D. W. (2002); *Applied Multivariate Statistical Analysis*, Pearson Education, Inc.

4) Johnson, Dallas E. (1998); *Applied Multivariate Methods for Data Analysis*, International Thomson Publishing Inc.

5) Kaiser, H.F. (1958); The Varimax criterion for analytical rotation in factor analysis, Psychometrica, 23, 187-200.

6) https://onlinecourses.science.psu.edu

7) https://en.wikipedia.org/wiki/Factor_Analysis

8) http://www.researchgate.net/file.PostFileLoader.html?id=53c50b3fd5a3f2140c8b465e&assetKey=AS%3A271747469774848%401441801053551.

13.14 ANSWERS OR HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) See Section 13.1

2) See Section 13.4

3) See Section 13.4

4) See Section 13.4

Check Your Progress 2

1) See Section 13.6

2) See Section 13.7

3) Hints: How to run Factor Analysis in SPSS?

After opening the SPSS windows sheet in your computer system, the following are the steps which can be used for performing, evaluating and analyzing the factor analysis solution:

Step 1 : Feed the entries in the SPSS work sheet.

Step 2 : Go to "Analyze" in the menu available at the top of the window screen.

Step 3 : Click on "Data Reduction"

Step 4 : Click on "Factor"

Step 5 : Click on "Extraction" and select "principal axis factoring" and press "Continue"

Step 6 : Click on "OK"

13.15 EXERCISES

1) What do you understand by the Factor score? Describe the different methods for the estimation of factor score.

2) Write a short note on Factor rotation and Oblique rotation.

3) What do you understand by the methods of estimation? Describe in details, the principal component and maximum likelihood methods.

4) With suitable example, explain communality in context to factor analysis.

5) Define the following terms:

   a) Correlation matrix

   b) Specific variance

   c) Factor loadings

   d) Specific factors